

Task Division Plan: Design Patterns (Group 52)

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Introduction

This document outlines a balanced, collaborative task division plan for two group members (Member A and Member B) working on the CS202 Design Patterns assignment.

To ensure shared learning and high-quality results, both members will actively cowork on the implementation, reports, and presentations of each pattern. Rather than dividing patterns individually (which isolates knowledge), the work is split functionally within each pattern.

Priority Matrix

- 1. **Observer Pattern (Priority 1 - High):** Full PPTX Presentation and LaTeX Report.
- 2. **Interpreter Pattern (Priority 2 - Medium):** LaTeX Report.
- 3. **Mediator Pattern (Priority 3 - Normal):** LaTeX Report.

Detailed Task Allocation

Phase 1: Observer Pattern (Priority 1)

Task	Member A	Member B	Milestone
C++ Code	Implement Naive YouTube Channel (<code>naive.cpp</code>).	Implement decoupled Observer patterns (<code>pattern.cpp</code>).	Compiles & runs in <code>main.cpp</code> .
UML Diagrams	Design specific class diagram for YouTube subscribers.	Design general class diagram for Observer pattern.	UML check & visual verify.
LaTeX Report	Draft sections 1–4 (Problem, Naive, Drawbacks, Intro).	Draft sections 7–9 (Implementation, Pros/Cons, Apps).	Compile report PDF.
PPTX Slides	Outline structure, draft slides 1–5 (Problem to Intro).	Format code, draft slides 6–10 (UML to Apps).	Slide design sync & rehearsal.
Interactive Quiz	Write questions 1–4 (ABCD).	Write questions 5–8 (ABCD).	Integrate quiz key.

Table 1: Observer Pattern Task Distribution

Phase 2: Interpreter Pattern (Priority 2)

Task	Member A	Member B	Milestone
C++ Code	Implement AST expressions in <code>pattern.cpp</code> .	Implement naive logic parsing in <code>naive.cpp</code> .	Verify nested expressions in driver.
UML Diagrams	Design general class diagram for Interpreter.	Design AST structure diagram for boolean parser.	AST layout verification.
LaTeX Report	Draft sections 7–9 (Implementation, Pros/Cons, Apps).	Draft sections 1–4 (Problem, Naive, Drawbacks, Intro).	Compile report PDF.
Interactive Quiz	Write questions 1–3 (ABCD).	Write questions 4–6 (ABCD).	Integrate quiz key.

Table 2: Interpreter Pattern Task Distribution

Phase 3: Mediator Pattern (Priority 3)

Task	Member A	Member B	Milestone
C++ Code	Implement naive flight lists tracking inside flights.	Implement central ATC-Tower and Flight colleague classes.	Verify runway clearing conflicts in driver.
UML Diagrams	Design specific class diagram for ATC Tower & Flights.	Design general class diagram for Mediator pattern.	Check visual linkage.
LaTeX Report	Draft sections 1–4 (Problem, Naive, Drawbacks, Intro).	Draft sections 7–9 (Implementation, Pros/Cons, Apps).	Compile report PDF.
Interactive Quiz	Write questions 4–6 (ABCD).	Write questions 1–3 (ABCD).	Integrate quiz key.

Table 3: Mediator Pattern Task Distribution

Coworking Rules

- **Frequent Commits:** Commit updates to git at the end of each sub-task to prevent huge merge conflicts.
- **Peer Review:** Review each other's C++ code files and LaTeX text paragraphs before merging into `main` / compilation.
- **Weekly Syncs:** Rehearse the presentation slides and review report files together before deadlines.